Start coding or generate with AI.

```
import pandas as pd

df_metadata = pd.read_csv('/content/metadata')
print("Metadata DataFrame loaded successfully. Displaying first 5 rows:")
df_metadata.head()
```

```
Metadata DataFrame loaded successfully. Displaying first 5 rows:
```

0	format: SequenceTagging
1	restricted_access: false
2	description: English named-entity recognition ...
3	name: ner-conll2003-eng

```
import spacy

spacy.cli.download("en_core_web_sm")
```

```
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

```
pip install spacy transformers torch pandas
```

```
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.13)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.10)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.3)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.5.2)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10)
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.3)
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.24.0)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (4.67.3)
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.2)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4)
Requirement already satisfied: pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spacy) (75.2.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (26.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.24.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3
Requirement already satisfied: shellingham in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->t
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spac
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->sp
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3
Requirement already satisfied: typer>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from typer-slim<1.0.0,>=0.3.0->sp
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.
```

```
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim<1.0
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim<1.0
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer>
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->type
```

```
import spacy
from spacy import displacy
from transformers import pipeline, AutoTokenizer, AutoModelForTokenClassification
import torch
import pandas
```

```
nlp = spacy.load("en_core_web_sm")
print("en_core_web_sm model loaded successfully.")
```

```
en_core_web_sm model loaded successfully.
```

```
sentences = [
    "The new AI model achieved a breakthrough in natural language understanding, demonstrating human-like conversational ab
    "In a thrilling match, the underdog team secured a stunning victory in the final seconds of the championship game.",
    "The government announced a new policy aimed at boosting economic growth and creating more job opportunities nationwide
    "Leading tech companies are investing heavily in quantum computing research, expecting it to revolutionize various indu
    "An international summit was held to address climate change concerns, with world leaders pledging stronger commitments
]

print("Generated sentences:")
for i, sentence in enumerate(sentences):
    print(f"{i+1}. {sentence}")
```

```
Generated sentences:
1. The new AI model achieved a breakthrough in natural language understanding, demonstrating human-like conversational abili
2. In a thrilling match, the underdog team secured a stunning victory in the final seconds of the championship game.
3. The government announced a new policy aimed at boosting economic growth and creating more job opportunities nationwide.
4. Leading tech companies are investing heavily in quantum computing research, expecting it to revolutionize various industr
5. An international summit was held to address climate change concerns, with world leaders pledging stronger commitments to
```

```
print("\n--- Named Entity Recognition Results ---")
for i, sentence in enumerate(sentences):
    print(f"\nSentence {i+1}: {sentence}")
    doc = nlp(sentence)
    if doc.ents:
        for ent in doc.ents:
            print(f"  - Text: {ent.text}, Label: {ent.label_}")
    else:
        print("  No entities found.")
```

```
--- Named Entity Recognition Results ---

Sentence 1: The new AI model achieved a breakthrough in natural language understanding, demonstrating human-like conversatio
  No entities found.

Sentence 2: In a thrilling match, the underdog team secured a stunning victory in the final seconds of the championship game
  - Text: the final seconds, Label: TIME

Sentence 3: The government announced a new policy aimed at boosting economic growth and creating more job opportunities nati
  No entities found.

Sentence 4: Leading tech companies are investing heavily in quantum computing research, expecting it to revolutionize variou
  No entities found.

Sentence 5: An international summit was held to address climate change concerns, with world leaders pledging stronger commit
  No entities found.
```

```
entity_data = []

for sentence_text in sentences:
    doc = nlp(sentence_text)
    if doc.ents:
        for ent in doc.ents:
            entity_data.append({
                'Sentence': sentence_text,
                'Entity': ent.text,
                'Label': ent.label_
            })
    else:
        # If no entities are found, record the sentence with empty entity/label
        entity_data.append({
            'Sentence': sentence_text,
            'Entity': 'No entities found',
            'Label': 'N/A'
```

```
        })

df_entities = pd.DataFrame(entity_data)

print("\n--- Detected Entities DataFrame ---")
display(df_entities)
```

```
--- Detected Entities DataFrame ---
```

	Sentence	Entity	Label
0	The new AI model achieved a breakthrough in na...	No entities found	N/A
1	In a thrilling match, the underdog team secure...	the final seconds	TIME
2	The government announced a new policy aimed at...	No entities found	N/A
3	Leading tech companies are investing heavily i...	No entities found	N/A
4	An international summit was held to address cl...	No entities found	N/A

```
print("\n--- Visualizing Named Entities ---")
for i, sentence in enumerate(sentences):
    doc = nlp(sentence)
    print(f"\nSentence {i+1}:")
    displacy.render(doc, style="ent", jupyter=True, options={
        "colors": {
            "TIME": "#ffcc00",
            "ORG": "#aa9cfc",
            "GPE": "#f0f080",
            "PERSON": "#e0a5a5",
            "NORP": "#9f7f9f",
            "LOC": "#bf8fbf",
            "PRODUCT": "#b5d5b5",
            "EVENT": "#9bd7e5",
            "WORK_OF_ART": "#ffc3a0",
            "LAW": "#b5cea8",
            "DATE": "#feca74",
            "PERCENT": "#e5cc7a",
            "MONEY": "#d5cd92",
            "QUANTITY": "#a0d4c0",
            "ORDINAL": "#a1b5c9",
            "CARDINAL": "#a1c9c9"
        }
    })
```

```
--- Visualizing Named Entities ---

Sentence 1:
/usr/local/lib/python3.12/dist-packages/spacy/displacy/__init__.py:215: UserWarning: [W006] No entities to visualize found i
  warnings.warn(Warnings.W006)
```

The new AI model achieved a breakthrough in natural language understanding, demonstrating human-like conversational abilities.

```
Sentence 2:
```

In a thrilling match, the underdog team secured a stunning victory in the final seconds TIME of the championship game.

```
Sentence 3:
```

The government announced a new policy aimed at boosting economic growth and creating more job opportunities nationwide.

```
Sentence 4:
```

Leading tech companies are investing heavily in quantum computing research, expecting it to revolutionize various industries.

```
Sentence 5:
```

An international summit was held to address climate change concerns, with world leaders pledging stronger commitments to sustainability.

```
# 1. Load tokenizer
tokenizer = AutoTokenizer.from_pretrained("dbmdz/bert-large-cased-finetuned-conll03-english")
print("Tokenizer loaded successfully.")

# 2. Load model
model = AutoModelForTokenClassification.from_pretrained("dbmdz/bert-large-cased-finetuned-conll03-english")
print("Model loaded successfully.")

# 3. Create NER pipeline
ner_pipeline = pipeline("ner", model=model, tokenizer=tokenizer)
print("NER pipeline created successfully.")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
config.json: 100%  998/998 [00:00<00:00, 77.3kB/s]
tokenizer_config.json: 100%  60.0/60.0 [00:00<00:00, 5.81kB/s]
vocab.txt: 213k/? [00:00<00:00, 9.70MB/s]
Tokenizer loaded successfully.
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
model.safetensors: 100%  1.33G/1.33G [00:25<00:00, 81.7MB/s]
Loading weights: 100%  391/391 [00:00<00:00, 624.21it/s, Materializing param=classifier.weight]
BertForTokenClassification LOAD REPORT from: dbmdz/bert-large-cased-finetuned-conll03-english
Key                      | Status     |  |
-------------------------+------------+--+-
bert.pooler.dense.weight | UNEXPECTED |  |
bert.pooler.dense.bias   | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Model loaded successfully.
NER pipeline created successfully.
```

```
print("\n--- Hugging Face Transformers NER Results ---")
for i, sentence in enumerate(sentences):
    print(f"\nSentence {i+1}: {sentence}")
    ner_results = ner_pipeline(sentence)
    if ner_results:
        for entity in ner_results:
            print(f"  - Word: {entity['word']}, Entity: {entity['entity']}, Score: {entity['score']:.4f}")
    else:
        print("  No entities found.")
```

```
--- Hugging Face Transformers NER Results ---

Sentence 1: The new AI model achieved a breakthrough in natural language understanding, demonstrating human-like conversatic
  - Word: AI, Entity: I-MISC, Score: 0.7770

Sentence 2: In a thrilling match, the underdog team secured a stunning victory in the final seconds of the championship game
  No entities found.

Sentence 3: The government announced a new policy aimed at boosting economic growth and creating more job opportunities nati
  No entities found.

Sentence 4: Leading tech companies are investing heavily in quantum computing research, expecting it to revolutionize variou
  No entities found.

Sentence 5: An international summit was held to address climate change concerns, with world leaders pledging stronger commit
  No entities found.
```

```
cleaned_hf_entity_data = []

for sentence_text in sentences:
    ner_results = ner_pipeline(sentence_text)
    if ner_results:
        for entity in ner_results:
            cleaned_hf_entity_data.append({
                'Sentence': sentence_text,
                'Entity': entity['word'], # 'word' field from the pipeline usually handles subword merging and '##' removal
                'Label': entity['entity'],
                'Confidence Score': entity['score']
            })
    else:
        cleaned_hf_entity_data.append({
            'Sentence': sentence_text,
            'Entity': 'No entities found',
            'Label': 'N/A',
            'Confidence Score': None
        })

df_hf_entities = pd.DataFrame(cleaned_hf_entity_data)

print("\n--- Cleaned Hugging Face Entities DataFrame ---")
display(df_hf_entities)
```

```
--- Cleaned Hugging Face Entities DataFrame ---
```

	Sentence	Entity	Label	Confidence Score
0	The new AI model achieved a breakthrough in na...	AI	I-MISC	0.776989
1	In a thrilling match, the underdog team secure...	No entities found	N/A	NaN
2	The government announced a new policy aimed at...	No entities found	N/A	NaN
3	Leading tech companies are investing heavily i...	No entities found	N/A	NaN
4	An international summit was held to address cl...	No entities found	N/A	NaN

Comparison of Named Entity Recognition Models

Feature	spaCy	Hugging Face Transformers
Model Type	Rule-based, statistical (e.g., CNN, BiLSTM)	Transformer-based (e.g., BERT)
Speed	Generally faster for common use cases	Can be slower due to model size
Accuracy (qualitative)	Good for general purpose NER, often faster to train	State-of-the-art accuracy, excellent context awareness
Context Handling	Good	Excellent (due to transformer architecture)
Confidence Score	Not directly available (rule-based or statistical probability)	Available (score per entity)

Identifying Incorrect or Missed Entity Predictions

By comparing the outputs from both spaCy and Hugging Face Transformers models, we can identify some instances where entities were either missed or interpreted differently.

Error/Missed Entity 1: spaCy missing "AI" (Sentence 1)

- **Sentence:** "The new AI model achieved a breakthrough in natural language understanding, demonstrating human-like conversational abilities."
- **spaCy Predicted Output:** No entities found.
- **Hugging Face Predicted Output:** `AI` as `I-MISC` (Score: 0.7770)
- **Explanation:** The `en_core_web_sm` spaCy model, while robust for general-purpose NER, might not be as adept at recognizing acronyms like "AI" as a specific entity type (like MISC - Miscellaneous). Its training data might treat "AI" as a general concept rather than a named entity. The transformer-based model from Hugging Face, being more context-aware and potentially trained on more diverse and recent text, was able to identify "AI" as a miscellaneous entity.

Error/Missed Entity 2: Hugging Face missing "the final seconds" (Sentence 2)

- **Sentence:** "In a thrilling match, the underdog team secured a stunning victory in the final seconds of the championship game."
- **spaCy Predicted Output:** `the final seconds` as `TIME`
- **Hugging Face Predicted Output:** No entities found.
- **Explanation:** This is an interesting case where spaCy's `en_core_web_sm` model, which often uses rule-based components and statistical models, successfully identified "the final seconds" as a `TIME` entity. In contrast, the Hugging Face model failed to tag this temporal expression. This could indicate a difference in how each model's training dataset defines or prioritizes `TIME` entities, or that the BERT model, fine-tuned on CoNLL-2003, might be more focused on entities like persons, organizations, and locations rather than general temporal phrases.

Error/Missed Entity 3: Both models missing "The government" as an organization (Sentence 3)

- **Sentence:** "The government announced a new policy aimed at boosting economic growth and creating more job opportunities nationwide."
- **spaCy Predicted Output:** No entities found.
- **Hugging Face Predicted Output:** No entities found.
- **Explanation:** Both models failed to recognize "The government" as an entity. While "government" is a common noun, in this context, "The government" acts as an organizational entity. General-purpose NER models often prioritize proper nouns (e.g., "Google", "United Nations") for `ORG` labels. Generic references to organizations, even when functioning as an entity in the sentence, might be overlooked if they don't strictly match the patterns of named entities in their training data. This highlights a limitation in recognizing implicit organizational entities.